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Could the Bank of England have avoided mis-forecasting UK inflation during 2021–24?☆

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ABSTRACT

The Bank of England badly mis-forecast UK annual consumer price inflation as it rose rapidly from 2021, prompting a review by Ben Bernanke. This raised many important issues, but other crucial problems were not addressed, as we discuss. Unpredictable shocks explain some of the bank's forecast failures, but tardy reactions also mattered. We show that successive large and increasing same-sign one-step-ahead forecast errors contain the information to estimate broken trends, applied to forecasting the UK's inflation over 2021–24. Compared with Bank of England projections, substantial gains in forecast performance can be made by rapidly detecting trend breaks and updating forecasting models when they occur.

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1. Introduction

A key stimulus for the Bank of England's review undertaken by Ben Bernanke (2024) was the bank's serious underestimation of UK inflation during 2021–2023 (see Figure 1 in Pill, 2023), although the Bank of England forecast errors were similar to those at some other central-banks. Moreover, Coroneo (2025) shows that despite their very large errors, the Bank of England's inflation forecasts were also no worse than those of standard benchmarks like a random walk and a scalar autoregression over this error-prone period. Consequently, whatever caused the bank's failures also affected widely used forecasting devices and hence were not uniquely due to the problems Bernanke identified. Nevertheless, the 12 recommendations in Bernanke's review do reveal worrying failings (see Aikman & Barwell, 2024, 2025 for discussion of both the Bank's performance and the review's recommendations). The Bank's response claims these issues will be fixed, but why did an agency in charge of monetary policy need advice to update and modernise its software and forecasting framework, manage and evaluate its models, and incorporate a more realistic representation of the monetary transmission mechanism, including both supply and demand forces? All of these seem essential, if minimal, requirements for useful policy, although we doubt those failings alone explain the forecast errors, as discussed below.

Hendry and Muellbauer (2018, 2024) highlight other key issues missing from Bernanke's report that the Bank must address when modelling and forecasting processes subject to breaks, like the recent UK inflation. First, as a major component of the Bank's modelling approach is a dynamic stochastic general equilibrium system (DSGE; see Bank of England, 1999, Smets & Wouters, 2003, and Burgess et al., 2013), account must be taken of the pernicious role of equilibrium-correction mechanisms (EqCMs) when forecasting after equilibrium-mean shifts, as they converge back to the old equilibrium irrespective of where the new equilibrium is (see Hendry, 2006). This

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criticism actually applies widely, as most econometric forecasting models are EqCMs (unless purely in differenced data, which brings other problems when levels are needed for policy); Martinez and Ericsson (2025) show the typical 'hedgehog' graphs that result after shifts. Worse, and a serious omission from Bernanke's report, such shifts invalidate the mathematical basis of DSGEs, and of the conventional theorems about conditional expectations which are then neither unbiased nor minimum mean square error (see Hendry & Mizon, 2014). Thus, DSGEs are not structural in the important sense of having constant representations, emphasising the need to test the invariance of models before a policy is implemented (see Castle, Hendry, & Martinez, 2017). Such model non-constancy probably exacerbated the bank's large and erratic forecasting errors during 2022 (see Fig. 3).

Extending the Bank's suite of models to include other formulations, like agent-based models (ABMs) as discussed by Armstrong, Hynes, and Kirman (2025) *inter alia*, would generate a wider range of policy views. Unfortunately, Bernanke's recommendation for reporting more scenarios in order to do so is unwise, as their requirements are equally stringent (see Hendry & Pretis, 2023). If the model being used to generate a scenario is non-constant, the scenario outcomes need not be a useful approximation to what will eventuate if the scenario is implemented.

Although there are specific challenges facing forecasting for monetary policy, including the conditional nature of the Bank's forecasts using market-determined interest rates facing feedbacks, fixing models immediately after they fail is far more difficult than modifying forecasts, and while both matter in a policy context, a clearer view of the future after a shift would help guide useful policies even when models 'fall-apart'. By definition, unpredictable events cannot be forecast, and their effects can only be observed when they later become in-sample. All empirical models are problematic for policy and forecasting after unanticipated distributional shifts, and the early 2020s suffered three identified by Castle, Doornik, and Hendry (2025). Unbridled climate change is likely to create many more, yet was another important issue missing from the Bernanke report.

The structure of the paper is as follows. Section 2 outlines the issues of forecasting after trend breaks, and then Section 3 briefly describes indicator saturation estimators. Section 4 explains the role of intercept corrections. Section 5 detects and then forecasts after sudden shifts in annual UK inflation over 2021(3)–2024(3): Section 5.1–Section 5.4 walk through the four major trend breaks leading to the combined forecasts over 2021(4)–2024(3) in Section 5.5. Section 6 compares those forecasts with Monetary Policy Report (MPR) projections. Section 7 concludes.

2. Implications for forecasting after breaks

As the bank's forecast failures were not unique to it nor to benchmark devices, we now discuss a potential solution to rapidly improving forecasts after shifts using univariate models, and compare those to the bank's forecasts in Section 6. Although by the late 1920s there was an understanding of the problems for modelling from unit roots (stochastic trends; see e.g., Yule, 1926 and Smith, 1926), and for forecasting facing unanticipated extraneous changes (see Smith, 1929), forecasters appear to have paid little attention to such developments and tended to assume that the processes under study were relatively stationary. Consequently, poor macroeconomic forecasts and forecast failures continued to occur, and (e.g. Klein, 1971) described using 'add factors' to correct recent forecast errors without investigating why they helped, although Suits (1957) had provided a perceptive discussion of dummy variables in regression.

Several successive large forecast errors could be due to outliers or mis-measurements, a step shift in the mean of the process, or a trend break. As Klein (1971) noted, a model can be 'put back on track' at the forecast origin by an impulse indicator (a dummy variable with a zero value except at that single observation) acting as an intercept correction (IC) because the IC equals the previous forecast error. As the next forecast then will be made by an unchanged model, an impulse indicator IC would actually worsen forecasts after a location shift. Clements and Hendry (1996) showed that a step indicator was needed to correct such forecast failures, but the parameter re-estimates of the model would then be changed. In turn, their IC would worsen forecasts after a trend break, which creates a sequence of increasing one-sided one-step-ahead forecast errors as the forecast origin advances. Consequently, a broken trend indicator would be required. But how could the new slope be rapidly and accurately estimated to avoid continuing forecast failure? This transpires to be feasible using very few forecast errors, and could have been discovered long ago had the information content of intercept corrections not remained largely unexplored.

When accurate forecasting is the objective, in addition to rapidly detecting a shift, a forecasting model for the post-shift period is required. To isolate the source of the succession of forecast errors, and so capture any sudden rapid shifts, we use a deterministic-trend model and test if the first two or three significant impulse indicators can be eliminated when replaced by a broken linear or log-linear trend, as well as rejecting a step shift. The theory was developed by Castle, Doornik, and Hendry (2024), and applied to forecasting UK inflation by Castle et al. (2025).

Using impulse indicators to correct forecast origin misforecasts has three advantages. First, since they act as one-off ICs, the next forecast commences from the forecast origin data value with unchanged parameter estimates. This will lead to a further similar forecast error if there is a location shift or trend break, but a larger error after an outlier or measurement error compared to not adding an IC. Second, successive large forecast errors reveal that the current model is inadequate and needs updating, but the few new observations available from which to do so are insufficient for reformulating estimated systems. However, extending a deterministic-trend model by a broken linear or log-linear trend is easily implemented. Finally, the adequacy of either of those trend extensions

can be tested by the resulting insignificance of the impulse indicators and an encompassing test against each other; see Castle et al. (2024) for Monte Carlo simulations. Because the new and old trends will diverge increasingly, ever larger forecast errors will result if not corrected. Consequently, despite having few (only two or sometimes three) post-break observations, the new trend can be estimated reasonably accurately and so continue to forecast adequately until another shift occurs, as Section 4 explains.

The two key features are that there is considerable information in large forecast errors, and although not previously noticed, ICs correctly capture the magnitude and form of that information (see Campos & Ericsson, 1999, on substantial information sustaining accurate estimates in small samples; and Castle et al., 2024, on how ICs reveal the size and type of shift, also explained below).

Since the (ex post) forecasts can be tested against the actual outcomes, we can evaluate the approach for the recent surge in UK inflation, taking the role of an investigator who sequentially forecasted one step ahead. On finding a sequence of large, same-signed one-step-ahead forecast errors despite correcting such errors using impulse indicators as ICs, we seek the earliest date each sequence of impulse indicators can be replaced by a broken linear or log-linear trend to improve forecasts when a sharp upswing is in progress. Using monthly time series over 2010(1)–2024(3) on the log of the UK consumer price index including owner occupiers' housing costs (CPIH), we find a series of sudden trend shifts, each of which can be detected in turn after a couple of large forecast errors, from which annual inflation forecasts can be derived. The forecasts are respectably accurate until the following shift, and one set is able to predict 16 months ahead across peak inflation and its ensuing slowdown in 2023.

3. Indicator saturation estimation

Indicator saturation estimators (ISEs) are designed to detect outliers, shifts in means, or breaks in trends (inter alia) at any point in a time series, despite not knowing their numbers, signs, magnitudes, or timings. Their role is to remove contaminating influences while retaining relevant explanatory variables to reveal the underlying relationship of interest, here the latest (and hopefully future) trend. The approach adds an indicator variable with the appropriate formulation for every observation in a sample of size T to the set of potential regressors, and then searches for significant indicators (see Hendry & Doornik, 2014). Indicator variables could be impulse indicators, $1_j = 1$ for $t = j$ and zero otherwise for $j = 1, \ldots, T$ (IIS; Hendry, Johansen, & Santos, 2008); step indicators, $S_j = 1_{t \le j}$ and zero otherwise (SIS; Castle, Doornik, Hendry, & Pretis, 2015); or trend indicators (TIS; see Walker, Pretis, Powell-Smith, & Goldacre, 2019), which are the cumulation of step indicators.

A tree search algorithm with expanding and contracting block searches allows all indicators to be investigated for possible significance; Castle, Doornik, and Hendry (2021) provide details of the search algorithm. Given the resulting high dimensionality, selection must use very tight significance levels α to control the probability of retaining irrelevant indicators, particularly if impulses, steps, and trends are searched jointly, denoted super-saturation (Ericsson, 2012). We set $\alpha = 0.0001$ for TIS ($\alpha = 0.005$ for SIS, and $\alpha = 0.01$ for IIS if needed), and select these sequentially using the results in Hendry and Johansen (2015), here for $T = 130$.

4. Explaining the role of intercept corrections

Let $\widehat{\epsilon}_{T+1|T} = y_{T+1} - \widehat{y}_{T+1|T}$ be the one-step-ahead forecast error at $T+1$ from the process $y_t = \boldsymbol{\delta}'\mathbf{x}_t + \epsilon_t$ by the model $\widehat{y}_t = \widehat{\boldsymbol{\delta}}'\mathbf{x}_t$ where the variance of ϵ_t is σ^2_ϵ. When x_t is a linear trend, the estimated variance of $\widehat{\delta}$ is $O(T^{-3})$, and so is relatively negligible. Hence, during locally stationary periods, $\widehat{\epsilon}_t = (\delta - \widehat{\delta})t + \epsilon_t$, where the variance of $\widehat{\epsilon}_t$ is approximately $\widehat{\sigma}^2_\epsilon$.

However, after an unanticipated trend shift at T, $y_{T+1} = \delta(T+1) + \delta^*(T+1-T) + \epsilon_{T+1}$ leads to forecast failure as $\widehat{\epsilon}_{T+1|T} = \delta^* + (\delta - \widehat{\delta})(T+1) + \epsilon_{T+1} \approx \delta^* + \epsilon_{T+1}$ (given the small variance of $\widehat{\delta}$) which will be large compared to $\widehat{\sigma}_\epsilon$. Consequently, $\widehat{\epsilon}_{T+1|T}$, and hence the IC data vector, $\mathsf{I}_{T+1} = \widehat{\epsilon}_{T+1|T} 1_{T+1} \approx \delta^* 1_{T+1}$, closely estimates the broken trend. Adding 1_{T+1} to the model before forecasting y_{T+2} entails that parameter estimates are unaltered, so the next one-step forecast error $\widehat{\epsilon}_{T+2|T+1} \approx \delta^*(T+2-T) + \epsilon_{T+2}$ will be same-signed and even larger, and so again dominated by the trend shift.

Including $\mathsf{I}_{T+2} = \widehat{\epsilon}_{T+2} 1_{T+2} \approx 2\delta^* 1_{T+2}$ as the next IC creates two significant ICs, namely $\delta^* 1_{T+1}$ and $2\delta^* 1_{T+2}$, so a broken trend starting at $T+1$ can be added for $t^* = 1, 2, \ldots$ and will estimate $\delta^* t^*$. When the data have actually experienced such a shift, the ICs should become insignificant and when eliminated, the linear or log-linear trend should be significant, so an encompassing test can decide between them (and a step shift, if that is a possibility). Castle et al. (2024) provide a formal analysis and simulation evidence. That ICs 'contain' mainly shifts when they correct significant forecast errors has been hiding in plain view, but underpins the surprising finding that two or three successive significant ICs can sustain accurate estimates of unknown shifts.

If the shift at $T+1$ had just been an outlier or measurement error, then $\widehat{\epsilon}_{T+2}$ would not be significantly large when no IC is used. If it was a step shift, a similar size of forecast error should result (i.e., δ^*, δ^*), whereas the next forecast error should be larger and same-signed if it was a new broken linear (as above) or log-linear trend (i.e., $\log(2)\delta^*$, $\log(3)\delta^*$). In the present setting of forecasting p_t by deterministic trends where previous trend shifts have been taken into account by TIS and so do not enter the forecast, the very fact that the forecast errors are significant entails they are large compared to $\widehat{\sigma}_\epsilon$.

5. Modelling and forecasting UK annual inflation

The succession of the Covid-19 pandemic lockdowns, supply chain disruption and the energy and food crises caused by Russia's invasion of Ukraine led to three rapid upswings in UK inflation. Here we investigate how quickly these could have been detected by modelling the log of

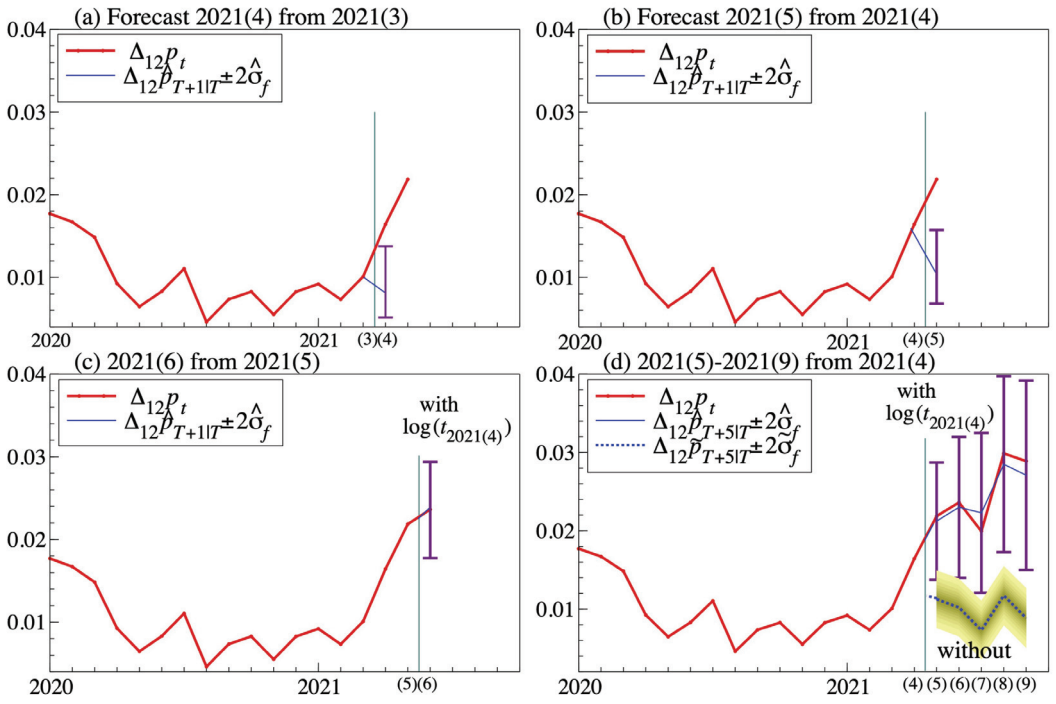


Fig. 1. Derived $\Delta_{12}p_t$ and forecast values $\widehat{\Delta_{12}p_t}$ showing the successive one-step-ahead forecasts for (a) 2021(4) from 2021(3), (b) 2021(5) from 2021(4), (c) 2021(6) from 2021(5) after adding $\log(t_{2021(4)})$ and dropping the insignificant ICs, and then (d) a multi-step forecast for 2021(9) from 2021(4) using $\log(t_{2021(4)})$ (dashed with error bars) and without (dotted with error fans). Months are shown in parentheses on the x-axis.

monthly consumer price index including owner occupiers' housing costs (CPIH), denoted p_t (source: Office of National Statistics) with a data set over 2010(1)–2024(3). We consider a forecaster making one-step-ahead forecasts as the forecast origin advances each month from 2021(3); this forecast could be the first of a multi-step sequence.

The initial model explains the log-level p_t by an intercept (denoted $\widehat{\beta}_0$) and linear trend ($\widehat{\beta}_1$), using TIS selected at $\alpha = 0.0001$ up to 2021(3) (see Castle et al., 2025, for detailed results). Ten trend shifts in the UK log price level between 2010 and 2021(3) were detected by TIS at 0.01%, correcting the overall estimated monthly trend to $\widehat{\beta}_1 = 0.071$ (i.e., 0.85% pa). Because all indicators are zero beyond their dates, the forecasts are simply $\widehat{p}_{T+h|T} = \widehat{\beta}_0 + \widehat{\beta}_1(T+h)$. Although we used p_t to test for and model changes in trend, the forecasts for annual inflation are easily derived from the log levels' forecasts as $\widehat{\Delta_{12}p}_{T+h|T} = \widehat{p}_{T+h|T} - p_{T+h-12}$, and will have the same error bars as the log levels, but re-centred on annual changes, as shown in Fig. 1.

After each of the first three unexpected breaks, the forecaster finds the first of their h-step forecasts is badly wrong, adds an IC, forecasts h-steps ahead again, and after a month discovers the next first step is also very badly wrong. Adding another IC for that error reveals two large and increasing errors, so tests to see which of a broken linear or log-linear trend eliminates them most effectively (essentially an encompassing test), then again forecasts h-steps ahead. So long as no large forecast errors occur, the forecaster continues with that selected model. If another break happens, the procedure just described is repeated. The graphs often show what would have happened if the break had been ignored to illustrate the power of the approach. Inadvertently, to see what the forecast would have been for the second month after the break using the broken trend, most of the multi-step forecasts shown below actually used a coefficient estimate based on just one large forecast error. Thus, for the final break, we selected a log-linear broken trend from one large forecast error, the theory of which is discussed in Castle et al. (2025).

5.1. The first trend break in 2021(4)

Fig. 1(a) shows the one-step-ahead forecast for 2021(4) from 2021(3) with $\mathsf{F}_{\mathsf{Chow}}(1, 123) = 10.5^{**}$ (see Chow, 1960), and Panel (b) records forecasting 2021(5) from 2021(4) after adding $\mathsf{I}_{2021(4)} = 0.0070$. The interval forecasts (denoted by bars reporting $\pm 2\widehat{\sigma}_f$) are not based on a congruent model, but offer a guide to the forecast uncertainty assuming no further trend breaks. When $\mathsf{I}_{2021(4)}$ is included, as in Panel (b), it acts as an intercept correction, so the next forecast commences from the 2021(4) outcome and hence leads to a large downward forecast as the upswing is in progress, emphasising the trend shift.

Define the broken trend $t_{2021(4)}$ as unity up to 2021(3), then 2, 3, 4, …from 2021(4) onwards. The damped version $\log(t_{2021(4)})$ is zero up to 2021(3) and follows a logarithmic trend afterwards. Adding $\log(t_{2021(4)})$ eliminates the two impulse indicators for 2021(4) and 2021(5), and has an insignificant Chow test. Panel (c) shows the result for forecasting 2021(6) from 2021(5).

Table 1
Forecast sequence leading to the broken trend $\log t_{2021(4)}$. '$T+1$ error' is the one-step-ahead forecast error. 'IC, TC added' shows the corrections to the baseline model, followed by the estimated coefficients of each added term.

T	$T+1$ error	IC,TC added	IC,TC coefficient(s)
2021(2)	0.002	–	–
2021(3)	0.007**	–	–
2021(4)	0.012**	$\mathsf{I}_{2021(4)}$	0.007**
2021(5)	0.015**	$\mathsf{I}_{2021(4)}$, $\mathsf{I}_{2021(5)}$	0.007**, 0.012**
2021(5)	−0.0002	$\mathsf{I}_{2021(4)}$, $\mathsf{I}_{2021(5)}$, $\log t_{2021(4)}$	−0.0004 ,–, 0.011**
2021(5)	0.00005	$\log t_{2021(4)}$	0.011**

** Significance at 1%.

Table 1 documents the steps leading to the inclusion of $\log(t_{2021(4)})$ as the forecast correction term.

Finally, Panel (d) starts estimation at 2021(4) and extends the forecast horizon to 2021(9). This yields a multi-step root mean-square forecast error, RMSFE = 0.0015 and $\mathsf{F}_{\mathsf{Chow}}(5, 123) = 0.52$, so actually forecasts better out of sample than the in-sample fit ($\widehat{\sigma} = 0.0020$), despite the coefficient of $\log(t_{2021(4)})$ being selected from just two observations and estimated from one. By way of contrast, the multi-step forecasts that would have occurred without adding $\log(t_{2021(4)})$ are shown dotted with fans.

The three out-of-sample forecasts from 2021(7) estimating to 2021(9) have an RMSFE of 0.0019, but the subsequent forecast fails. A similar procedure is applied for forecasting after the next three trend breaks, namely adding a new broken trend based on two successive ICs, as described in Section 5.2–5.4 with an overview in Section 5.5; see Castle et al. (2025) for further details.

5.2. The next trend break in 2021(10)

When making the broken log-linear trend forecast from 2021(7), the forecaster cannot know how far ahead forecasts will be reliable. The forecaster thus continues to make multi-step-ahead forecasts as the forecast origin advances through 2021(9), until forecasting 2021(10) delivers $\mathsf{F}_{\mathsf{Chow}}(1, 128) = 12.38^{**}$ and so a second break has happened, confirmed when forecasting 2021(11) from 2021(10), which suggested the break had started in 2021(8). Following a similar procedure, a broken linear trend starting in 2021(9) made the two ICs for 2021(10) and 2021(11) insignificant with an insignificant Chow test. Multi-step forecasts from 2021(11) to 2022(3) had $\mathsf{F}_{\mathsf{Chow}}(5, 128) = 1.81$ and a RMSFE = 0.4%.

5.3. The third trend break in 2022(4)

However, forecasting 2022(4) from 2022(3) leads to another significant failure with $\mathsf{F}_{\mathsf{Chow}}(1, 133) = 63^{**}$. By the time this shift could have been observed, Russia's invasion of Ukraine and the consequent energy crisis and fuel and food price rises had occurred, so such a shift would not be a surprise, confirmed by a large error forecasting 2022(5) from 2022(4).

As before, adding the next broken log-linear trend $\log(t_{2022(4)})$ delivers $\mathsf{F}_{\mathsf{Chow}}(1, 133) = 0.04$ for 2022(5) and eliminates the two ICs at 1%. The model with $\log(t_{2022(3)})$ continues to forecast reasonably accurately through to 2023(9), which is 17 periods ahead (see Figs. 2 for p_t and 3 for $\Delta_{12} p_t$), of which **16** are after selection from just two large forecast errors, confirming that medium-term forecasts can be usefully accurate despite the most recent broken trend being estimated from just one observation. Moreover, these multi-step forecast over 2022(5)–2023(9) show that despite using three broken trends with positive coefficients, the approach can forecast annual inflation peaking and then eight falls, reflecting that the previous year's inflation was higher than the current one.

5.4. A final shift in 2023(10)

The error forecasting 2023(9) for 2023(8) is insignificant, but when forecasting 2023(10) from 2023(9), a significant forecast error occurs with $\mathsf{F}_{\mathsf{Chow}}(1, 150) = 7.3^{*}$ revealing another trend break as the increases in p_t slowed. Once again, a broken log-linear trend from 2023(10) produces $\mathsf{F}_{\mathsf{Chow}}(5, 150) = 1.1$ over 2023(11)–2024(3) although selected from and fitted to just one large forecast error. That broken trend coefficient estimate is −0.0105 (0.0039), and estimated ex post until 2024(3) is −0.011 (0.0013); hence the accurate forecasts.

5.5. The combined forecasts over 2021(4)–2024(3)

Although we have described and evaluated the forecasts in terms of annual inflation, the procedure was actually applied to forecasting p_t, from which inflation forecasts were derived. Fig. 2 collects all our multi-step-ahead monthly forecasts and outcomes for p_t over 2021(4)–2024(3), including the sequences of large forecast errors that prompted the additions of broken trends, highlighted by boxes.

The four essentially unpredictable trend shifts are clearly visible in Fig. 2, and were followed by a couple of significant forecast errors. But overall we only experienced seven large errors out of 36 forecasts when rapidly detecting breaks, rather than long periods of systematic forecast failure. Relative to our RMSFEs in Table 2 below, the interval forecasts turned out to be too narrow before correcting for shifts, and too wide after. Overall, the approach of adding ICs after observed one-step forecast failures and then replacing these by broken trends provided greatly improved forecasts. We now consider how these compared with the Bank of England's forecasts over 2021(4)–2024(3).

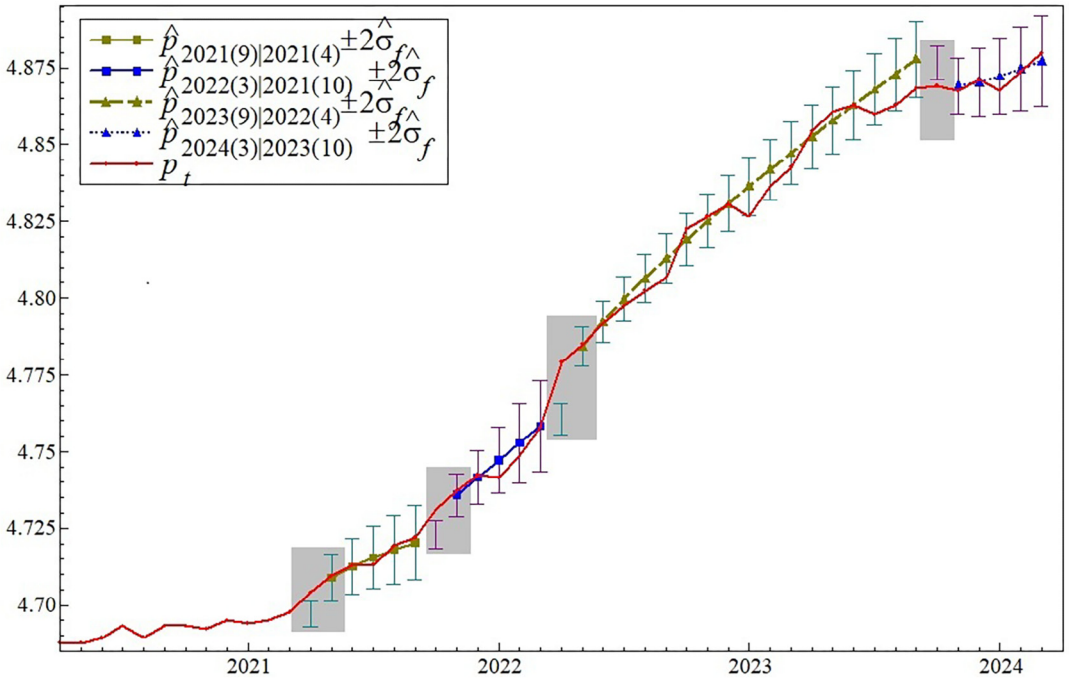


Fig. 2. Multi-step-ahead forecasts for $\widehat{p}_{T+h|T}$ (dashed) with outcomes p_{T+h} (solid) over the inflation upsurge and fall from 2021(4)–2024(3), spanning the four break episodes. The boxes highlight the breaks where the first error bar shows the initial mis-forecast but the second is based on the 1-observation broken trend estimate.

6. Comparisons with monetary policy report projections

The Bernanke review of modelling and forecasting at the Bank of England raised a number of issues about the bank's recent poor forecasting performance, clearly visible in Fig. 3, which plots the Monetary Policy Report (MPR) projections of annual inflation rates over the crisis period, as well as ours.

The Bank started raising interest rates from 0.1% in February 2022, continuing to raise in small steps until 5.25% in August 2023, yet our accurate 16-month-ahead forecasts to 2023(9) were made in 2022(4) prior to most of those increases. Although our forecasts were obviously made afterwards, they were only based on information available at the time. Had the method in Castle et al. (2024) been developed before 2022(4), the same forecasts would have been made; one wonders what the Monetary Policy Committee (MPC) will make of them. Possible implications of the forecasts ignoring the main tool of monetary policy range from 'the impacts of interest rates on inflation are small' and 'the inflationary surge was mainly global energy price driven, so domestic policy had little effect' to 'interest rate increases slowed a linear trend price increase to a log-linear, happening to match our forecasts'. If the bank is to continue basing monetary policy on interest rates, it needs to ascertain why our deterministic trends could forecast so accurately. Hendry and Muellbauer (2024) estimated the impacts of import price inflation, energy price rises of 170%, and fuel shortages on UK inflation in 2022, calculating that they accounted for about three-quarters of the peak 9% rise in CPIH. When all those external inflationary influences dropped considerably in 2023, UK inflation rose more slowly, so the second conjecture seems most likely. Nevertheless, the wage-price spiral results in Castle and Hendry (2014) are consistent with the increased interest rates dampening later potential second-round effects. Moreover, our method is designed to rapidly detect trend breaks, usually due to extraneous sources, and was clearly successful at doing so in practice, so the MPC could incorporate it as another approach in its suite of models.

Our approach was developed to detect rapid upsurges as early as possible to avoid continuing large forecast failures, rather than forecast from given dates, and is only relevant once a shift has led to large one-step forecast errors. The first such shift was found in 2021(4) following the end of lockdowns with the amazingly quick availability of vaccines, and confirmed the forecast for 2021(6), enabling $\log(t_{2021(4)})$ to be estimated and forecast until 2021(9) when a second shift was detected in 2021(10). The MPR projection in February 2021 would not know at that time of either shift, and in May could not know of the 2021(10) shift from the supply-chain disruptions; hence their inaccuracy. Once we detected that shift, our end-year forecast was very accurate, as shown in Fig. 3, as was the final MPR's. Similarly, the next shift in 2022(3) was just after the MPR's first year-end projection, so it was seriously wrong, but detecting it by 2022(5), our approach provided a much more accurate end-year forecast relative to the MPR's remaining dates (indeed through to 2023(9)),

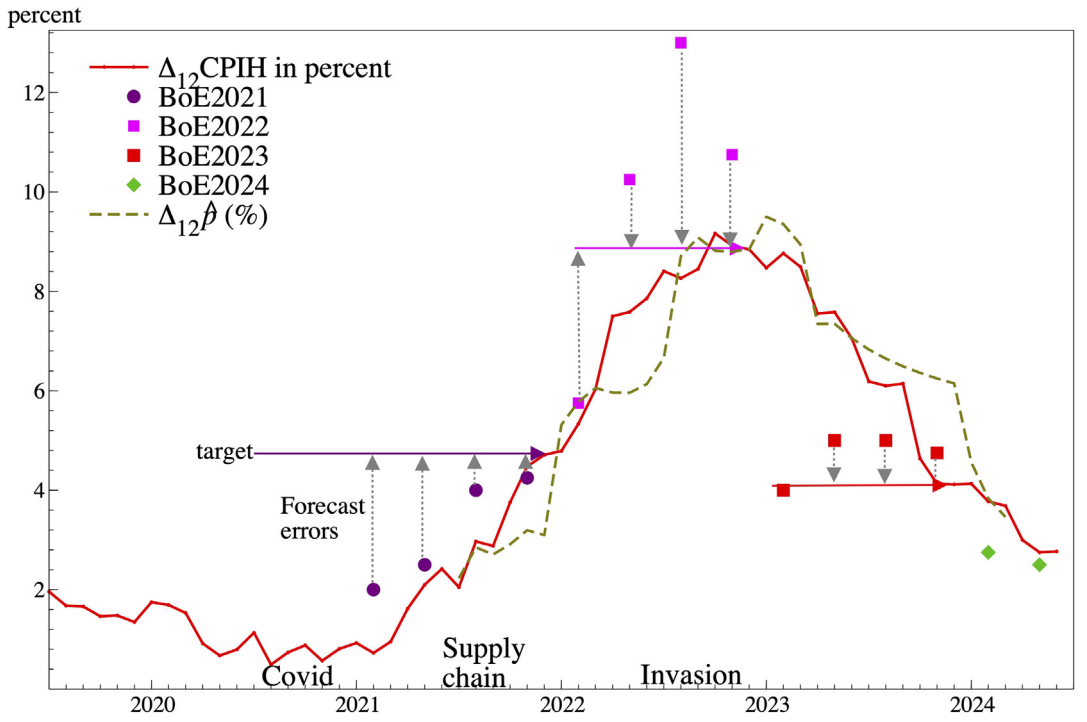


Fig. 3. The four-times-a-year projections in the MPR for each December are shown as symbols, and the forecasts denoted $\Delta_{12}\widehat{p}$ from our approach (dashed). The three horizontal lines link the four MPC forecasts (made February, May, August, and November) to the target outcome of the annual inflation rate in December. The bank's forecast errors are the deviations of the points to those lines, indicated by the vertical arrows; all numbers are percentages.

Table 2
Root-mean-square forecast errors in percent.

Source	CDH	BoE	BoE
horizon months	$\overline{3}$	3	6
RMSFE (%)	0.67	0.93	1.66

avoiding the MPR's erratic over-shoots.[1] Conversely, our forecast of the December 2023 inflation rate from May was less accurate as the shift occurred in 2023(10). Although the bank's timing was unlucky, making forecasts at fixed intervals seems suboptimal compared to producing them at every feasible forecast origin when shifts are likely to occur.

Our forecasts in Fig. 3 are pseudo-out-of-sample. For each break period that was used to update the forecasts, we include the two significantly wrong forecasts. Only afterwards do we switch to the new forecasts. The data and PcGive batch file can be downloaded.

Table 2 records our RMSFEs (denoted CDH) and compares these to the Bank of England (BoE) quarterly forecast errors all in percentages, based on the second subsample 2019(Q1)–2023(Q4) in Coroneo (2025, Table 1) which includes two years of relatively quiescent inflation. Our RMSFEs are calculated from the monthly multi-step ahead forecast errors for $\Delta_{12}p_t$ over 2021(4)–2024(3), as shown in Fig. 3. Those for the BoE quarterly forecast errors are reported both at approximately the same average step ahead horizon of 3 months and the longer 6 months. Our average horizon is 3 from (1,1,4,1,1,4,1,1,16,1,5) step-ahead forecasts respectively, so the BoE three-month-ahead forecast RMSFEs are much larger. Given such significant gains, continual monitoring and forecasting alongside fixed interval forecasts should help to avoid delays in responding to trend breaks in inflation. As soon as breaks in trend are detected in real time, analysts should use their information to investigate the underlying causes, as described in Section 4, and update their forecasting models accordingly, also helping to avoid the models adjusting to the incorrect past trend. Other than immediately after each unexpected shift, our forecasts are very accurate, and so to help avoid future systematic forecast failure from unmodeled unexpected shifts, the Bank could usefully add this approach to their suite of models: 'a trend is your friend till it doth bend', which it has done all too frequently over the last few years, but can now be rapidly corrected.

7. Conclusion

That economic forecasting is fundamentally different from modelling, and need not be based on empirical models, was not addressed by Bernanke. Forecasts can be guesses, extrapolations, leading indicators, or model-based, and may be averages of several predictors. Moreover, a given model can produce a variety of very different

[1] Technically, forecasts of annual inflation more than 12 months ahead should be calculated as $\Delta_{12}\widehat{p}_{T+h|T} = \widehat{p}_{T+h|T} - \widehat{p}_{T+h-12}$ for $h > 12$, but the five forecasts of p that would be involved are very accurate.

forecasts depending on whether ICs are used or not (and what type), how 'offline' variables like interest rates are forecast, and whether other robustification methods were used (see e.g., Martinez, Castle, & Hendry, 2022). The major breaks from Covid-19 lockdowns followed by the extraordinarily rapid development of vaccines, supply chain disruptions, and the energy crisis after the Russian invasion of Ukraine are recent large unanticipated shocks that would cause failures in forecasts made before they were known, even if the model used was the prior data generation process. The best possible model need not produce accurate forecasts facing sudden structural shifts. Major problems confront forecasting from economic models, in addition to the difficulties facing EqCMs after shifts: Hendry and Mizon (2012) delineate 19 potential forecast error sources for conditional models, relevant to the Bank's forecasts using market-determined interest rates rather than their own model-consistent forecasts. It is a stretch to describe the deterministic trend device here as a 'good model', yet using the information content of successions of two large forecast errors, it greatly outperforms the Bank's suite of model forecasts.

Social and economic processes are always changing, often abruptly and unexpectedly. Large shifts will lead to inaccurate forecasts and changes in estimated parameters of empirical models. Our developments of indicator saturation estimators, like TIS, are designed to reveal any constant relationships camouflaged by such shifts, albeit future shifts are bound to occur. It is difficult to quickly discern the causes of large forecast errors and harder still to rapidly implement structural changes to econometric systems. Our procedure provides a solution to rapidly detecting shifts to improve forecasts but is silent on what changes may be entailed for models. As Castle et al. (2024) demonstrate for the huge surge in narrow money demand after 1984Q4, it is also applicable to correcting forecasts from aberrant equations in econometric systems prior to implementing an explanation as in Hendry and Ericsson (1991).

While many of Ben Bernanke's 12 recommendations were sorely needed, others were not well thought out, and at least equally important recommendations were missing, as we discussed in the introduction. Nevertheless, his review paves the way for much needed changes, but its omissions about how to forecast are equally revealing and also need to be addressed by the Bank to improve its forecast performance facing repeated breaks.

CRediT authorship contribution statement

Jennifer L. Castle: Writing – original draft, Methodology, Formal analysis, Conceptualization, Writing – review & editing, Validation, Investigation, Data curation. **Jurgen A. Doornik:** Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization, Writing – review & editing, Visualization, Software, Investigation, Data curation. **David F. Hendry:** Writing – original draft, Validation, Investigation, Formal analysis, Conceptualization, Writing – review & editing, Visualization, Methodology, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: David Hendry reports financial support was provided by University of Oxford Nuffield College. Jurgen Doornik reports a relationship with University of Oxford Nuffield College that includes: employment. Royalties are received from the software used.

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Data and code availability

All data and code necessary to reproduce the results gathered in the manuscript are available at https://www.climateeconometrics.org/the-bernanke-review-overview-and-data.

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